Mathematics: analysis and approaches
Standard level
Paper 1 — markscheme
Topic 4 Statistics — box-and-whisker diagrams

5 November 2026

W4ss Academy practice set 1 | **Topic** Topic 4 Statistics — box-and-whisker diagrams | **Ref.** W4SS–MAA–QB–P1–01

Student name

Paper 1

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- Total: **[24 marks]**.

W4ss Academy 2026

Section A

Question	Marks	Markscheme
1.	**[2]**	Cafeteria 2 ***(A1)(C1)*** 75 % > 50 % (do not meet the requirement) ***(R1)(C1)*** **OR** 25 % < 50 % (meet the requirement) ***(R1)(C1)*** **Note:** Do not award (A1)(R0). Award the (R1) for a correct comparison of percentages for both cafeterias, which may be in words. The percentage values or fractions must be seen. It is possible to award (A0)(R1).
2.	**[1]**	125 (accept 125.5) ***(A1)***
3. (a)	**[1]**	42 (minutes) ***(A1)(C1)***
3. (b)	**[2]**	Stephen is correct. ***(A1)*** the minimum rehearsal time is greater than zero ***(R1)*** **OR** he rehearsed at least 20 minutes every day ***(R1)(C2)*** **Note:** Do not award (A1)(R0). Accept equivalent reasoning based on the box-and-whisker diagram.
3. (c)	**[3]**	0, 2, 3, 4, …, 15 ***(A1)(A1)(A1)(C3)*** **Note:** Award (A1)(A1) for each correct endpoint of the interval, (A1) for indication of integer values, except 1, between their endpoints.

Section B

Question	Marks	Markscheme
4. (a)	**[2]**	valid approach ***(M1)*** *eg* $Q_3 - Q_1$, $Q_3 - 1.1$, $4.5 - a = 1.1$ $a = 3.4$ ***A1 N2***
4. (b)	**[1]**	$\frac{32}{5}$ (= 6.4) (km) ***A1 N1***

4. (c)	**[4]**	**METHOD 1 (standard deviation first)** valid approach ***(M1)*** *eg* standard deviation = $\sqrt{\text{variance}}$, $\sqrt{\frac{16}{9}}$ standard deviation = $\frac{4}{3}$ (km) ***(A1)*** valid approach to convert **their** standard deviation ***(M1)*** *eg* $\frac{4}{3} \times \frac{5}{8}$, $\sqrt{\frac{16}{9}} = \frac{8}{5}M$ $\frac{20}{24}$ (miles) $(= \frac{5}{6})$ ***A1 N3*** **Note:** If no working shown, award ***M1A1M0A0*** for the value $\frac{4}{3}$. If working shown, and candidate's final answer is $\frac{4}{3}$, award ***M1A1M0A0***. **METHOD 2 (variance first)** valid approach to convert variance ***(M1)*** *eg* $(\frac{5}{8})^2$, $\frac{64}{25}$, $\frac{16}{9} \times (\frac{5}{8})^2$ variance $= \frac{25}{36}$ ***(A1)*** valid approach ***(M1)*** *eg* standard deviation = $\sqrt{\text{variance}}$, $\sqrt{\frac{25}{36}}$, $\sqrt{\frac{16}{9} \times (\frac{5}{8})^2}$ $\frac{20}{24}$ (miles) $(= \frac{5}{6})$ ***A1 N3***
4. (d)	**[3]**	correct frequency for 22 minutes ***(A1)*** *eg* 20 adding **their** frequency (do not accept 22 + 400) ***(M1)*** *eg* 20 + 400, 420 athletes m = 30 (minutes) ***A1 N3***

4. (e)	**[5]**	27 (minutes) ***(A1)*** correct working ***(A1)*** *eg* 130 athletes between 22 and 27 minutes, $P(22 < t < 27) = \frac{150 - 20}{600}$, $\frac{13}{60}$ evidence of conditional probability or reduced sample space ***(M1)*** *eg* $P(A \mid B)$, $P(t < 27 \mid 22 < t < 30)$, $\frac{P(22 < t < 27)}{P(22 < t < m)}$, $\frac{150}{400}$ correct working ***(A1)*** *eg* $\frac{\frac{130}{600}}{\frac{400}{600}}$, $\frac{150 - 20}{400}$ $\frac{130}{400}$ ($\frac{13}{40} = \frac{78000}{240000} = \frac{390}{1200} = 0.325$) ***A1*** ***N5*** **Note:** If no other working is shown, award ***A0A0M1A0A0*** for answer of $\frac{150}{400}$. Award ***N0*** for answer of $\frac{3}{8}$ with no other working shown.